COUNTING METHODS
AND
PIGEONHOLE PRINCIPLE

Basic Principles of Counting

Multiplication Principle

- Theorem:

  If an operation can be performed in n_1 ways, and if for each of these a second operation can be performed in n_2 ways, and for each of the first two a third operation can be performed in n_3 ways, etc., then the sequence of k operations can be performed in $n_1 \bullet n_2 \bullet n_3 \bullet \ldots \bullet n_k$ ways.

- Remark :

  This principle is also known as The Fundamental Principle of Counting

Multiplication Principle

Example 1:

How many lunches are possible consisting of soup, a sandwich, dessert and a drink if one can select from 4 soups, 3 kinds of sandwiches, 5 desserts and 4 drinks?

Solution :

The total number of lunches possible would be $4 \bullet 3 \bullet 5 \bullet 4 = 240$

Multiplication Principle

Example 2:

How many even three-digit numbers can be formed from the digits 1, 2, 5, 6, 9 if each digit can be used only once?

Solution :

Since the number is even, we have only 2 choices for the units position. For each of these, we have 4 choices for the tens position and three choices for the hundreds position. Therefore we can form a total of (2)(4)(3) = 24 even three- digit numbers.

Multiplication Principle

Example 3:

How many values are possible for an eight-bit string?

Solution :

An eight-bit encoding can be constructed in eight successive steps: Select the first bit, select the second bit and so on up to the eighth bit. Since there are two ways to select each bit, by the Multiplication Principle the total number of eight-bit encoding is

$2 \bullet 2 \bullet 2 \bullet 2 \bullet 2 \bullet 2 \bullet 2 \bullet 2 = 2^8 = 256$

Addition Principle

Theorem :

Suppose that $X_1, \ldots, X_k$ are sets and that the ith set X_i has n_i elements. If $\{X_1, \ldots, X_k\}$ is a pairwise disjoint family (i.e., if $i \neq j$, $X_i \cap X_j = \varnothing$), the number of possible elements that can be selected from X_1 or X_2 or X_k is

$$n_1 + n_2 + \ldots + n_k$$

(Equivalently, the union $X_1 \cup X_2 \cup \ldots \cup X_t$ contains $n_1 + n_2 + \ldots + n_k$ elements.)

Addition Principle

Example 4:

How many eight-bit strings with either 101 or 111?

Solution:

An eight-bit string that begins with 101 can be constructed in five successive steps: select the fourth bit, select the fifth bit, and so on up to eighth bit. Since each of the five bits can be selected in tow ways, by Multiplication Principle, there are

$$2 \bullet 2 \bullet 2 \bullet 2 \bullet 2 = 2^5 = 32$$

eight-bit strings that begin with 101.

Addition Principle

Example 4:

How many eight-bit strings with either 101 or 111?

Solution (*cont.*) :

The same argument can be used to show that there are 32 eight-bit strings that begin with 111. Since there are 32 eight-bit strings that begin with 101 and 32 eight-bit strings that begin with 111, there are 32 + 32 = 64 eight-bit-strings that begin either 101 or 111.

In this example, we added the numbers of eight-bit strings (32 and 32) of each type to determine the final result.

Example 5:

A six-person committee composed of Alice, Ben, Connie, Dolph, Egbert, and Francisco is to select a chairperson, secretary, and treasurer.

In how many ways can this be done?

Solution :

We use the Multiplication Principle. The officers can be selected in three successive steps: Select the chairperson, select the secretary, select the treasurer. The chairperson can be selected in six ways. Once the chairperson has been selected, the secretary can be selected in five ways. After selection of the chairperson and secretary, the treasurer can be selected in four ways. Therefore, the total number of possibilities is

$$6 \,.\, 5 \,.\, 4 = 120$$

Permutations and Combinations

Permutation

Definition :

A *permutation* is an arrangement of all or part of a set of objects.

Permutation

Example 1 :

Consider the three letters **a**, **b**, and **c**. The possible permutations are **abc**, **acb**, **bac**, **bca**, **cab**, and **cba**. Thus we see that there are 6 distinct arrangements.

Solutions:

Using the Multiplication Principle, we could have arrived at the answer without actually listing the different orders. There are 3 positions to be filled from the letters a, b, and c. Therefore, we have 3 choices for the first position, and 2 for the second, leaving only 1 choice for the last position, giving a total of (3)(2)(1) = 6 permutations.

In general n distinct objects can be arranged in n(n - 1)(n - 2) . . . (3)(2)(1) ways. We represent this product by the symbol n!, which is read "n factorial." Three objects can be arranged in 3! = (3)(2)(1) = 6 ways. By definition 1! = 1 and 0! = 1.

Permutation

Theorem:

The number of permutations of **n** distinct objects in **n!**.

Proof:

We use the Multiplication Principle. A permutation of n elements can be constructed in n successive steps: Select the first element, select the second element, . . . , select the last element. The first element can be selected in n ways. Once the first element has been selected, the second element can be. selected in n - 1 ways. Once the second element has been selected, the third element can be selected in n - 2 ways, and so on. By the by the Multiplication Principle, there are

$$n(n - 1)(n - 2). . .2. 1 = n!$$

permutations of n elements.

Permutation

Example 2:

The number of permutations of the four letters *a*, *b*, *c*, and *d* will be 4! = 24.

Permutation

Definition:

An *r-permutation* of **n** (distinct) elements $X_1 \ldots X_n$ is an ordering of an r-element subset of $\{ X_1 \ . \ . \ X_n\}$, or the number of permutations of **n** distinct object taken **r** at a time. The number of r-permutations of a set of n distinct elements is denoted P(n, r).

Permutation

Theorem:

The number of r-permutations of a set of n distinct objects is

$$P(n, r) = n(n-1)(n-2) \ldots (n-r+1), \qquad r \leq n.$$

We represent this product by the symbol

$${}_nP_r = \frac{n!}{(n-r)!}.$$

Proof:

We are to count the number of ways to order r elements selected from an n-element set. The first element can be selected in n ways. Once the first element has been selected, the second element can be selected in n - 1 ways. We continue selecting elements until we have selected the (r - 1)th element, we select the rth element. This last element can be chosen in n - r + 1 ways. By the Multiplication Principle, the number of r-permutations of a set of n distinct objects is

$$n(n-1)(n-2)\ldots(n-r+.1)$$

Permutation

Example 3:

Consider the number of permutations that are possible by taking the 4 letters **a**, **b**, **c**, and **d**, 2 at a time. These would be **ab**, **ac**, **ad**, **ba**, **ca**, **da**, **bc**, **cb**, **bd**, **db**, **cd**, and **dc**. Hence, the number of permutations possible is 12.

Solution:

Notice that this situation is actually a 2-permutation problem on 4 objects, or permutation of 4 objects taken 2 at a time. Applying the above theorem, we have

$$ {}_4P_2 = \frac{4!}{(4-2)!} = \frac{4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 1} = 12. $$

Permutation

Example 4:

Find the number of ways that two lottery tickets can be drawn from 20 tickets for first and second prizes.

Solution:

The total number of possibilities is

$$_{20}P_2 = \frac{20!}{(20-2)!} = \frac{20!}{18!} = 20 \cdot 19 = 380$$

Permutation

Example 5:

How many ways can a basketball team schedule 3 exhibition games with 3 teams if they are all available on any of 5 possible dates?

Solution:

The total number of possible schedules is

$$ {}_5P_3 = \frac{5!}{(5-3)!} = \frac{5!}{2!} = 5 \cdot 4 \cdot 3 = 60. $$

Permutation

Definition:

Permutations that occur by arranging objects in a circle are called *circular permutations*.

Remark:

Two circular permutations are not considered different, unless corresponding objects in the two arrangements are preceded or followed by a different objects as we proceed in a clockwise direction.

Permutation

Theorem:

The number of circular permutations of a set of n distinct objects is

$$(n-1)! = (n-1)(n-2)\ldots 3 \cdot 2 \cdot 1$$

Proof:

For linear permutation, that is, arranging **n** objects in a linear fashion, we know that we have **n!** possible ways. From the discussion above, we know that two circular permutations are not considered different, that is, in moving all objects one position in a clockwise or counter-clockwise direction, no new circular permutation is formed.

In a linear permutation, this movement would have resulted to a new arrangement. Since, there are **n** objects, then **n** positions are possible for the movement. Thus, we take this factor out **n!**, leaving us with **(n –1)!** possibilities for circular permutation.

Permutation

Theorem:

The number of distinct permutations of **n** objects of which $\mathbf{n_1}$ are of one kind, $\mathbf{n_2}$ are of second kind, … , $\mathbf{n_k}$ are of a **kth** kind is,

$$\frac{n!}{n_1!n_2\cdots n_k!}$$

Permutation

Example 6:

How many different ways can 3 red, 4 yellow, and 2 blue bulbs be arranged in a string of Christmas tree lights with 9 sockets?

Solution:

The total number of distinct arrangements is

$$\frac{9!}{3!\cdot 4!\cdot 2!} = 1260$$

Permutation

Theorem:

The number of ways of partitioning a set of n objects into r cells with n1 elements in the first cell, n2 elements in the second, and so on is

$$\binom{n}{n_1, n_2, \ldots n_k} = \frac{n!}{n_1! n_2! \cdots n_k!},$$

where $n_1 + n_2 + \ldots + n_r = n$

Remark:

The above theorem is used if we are interested in the number of ways of partitioning a set of n objects into r subsets, called cells. A partition has been achieved if the intersection of every possible pair of the r subsets is the empty set and the union of all subsets gives the original set. The order of the elements within a cell is of no importance

Permutation

Example 7:

Consider the set {a,e,i,o,u}. Find the number of possible partitions of this set into 2 cells, such that the first cell contains 4 elements and the second cell 1 element.

Solution:

The possible partitions of the set {a,e,i,o,u} into 2 cells, such that the first cell contains 4 elements and the second cell 1 element are {(a, e, i, o), (u)}, {(a, i, o, u), (e)}, {(e, i, o, u), (a)}, {(a, e, o, u), (i)}. We see that there are 5 such ways to partition a set of 5 elements into two subsets or cells containing 4 elements in the first cell and 1 element in the second. For this example the number of partitions is denoted by

$$\binom{5}{4,\ 1} = \frac{5!}{4!\ 1!}, = 5,$$

where the top number represents the total number of elements and the bottom numbers represent the number of elements going into each cell.

Permutation

Example 8:

How many ways can 7 people be assigned to 1 triple and 2 double rooms?

Solution:

The total number of possible partitions would be

$$\binom{7}{3,\ 2,\ 2} = \frac{7!}{3!\ 2!\ 2!}, = 210$$

Combination

Definition:

A *combination* in the number of ways of selecting **r** objects from **n** objects without regard to order.

Theorem:

The number of combination of **n** distinct objects taken **r** at a time is

$$_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Combination

Example 9:

From 4 girls and 3 boys find the number of team of 3 that can be formed with 2 girls and 1 boy.

Solution:

The number of ways of selecting 2 girls from 4 is $\binom{4}{2} = \frac{4!}{2!\ 2!}, = 6$

The number of ways of selecting 1 boy from 3 is $\binom{3}{1} = \frac{3!}{1!\ 2!}, = 3$

Using the multiplication principle, we find that the number of teams that can be formed from 4 girls and 3 boys with 2 girls and 1 boy to be (6)(3) = 18.

The Pigeonhole Principle

Pigeonhole Principle

The *Pigeonhole Principle* also known as the Dirichlet Drawer Principle or the Shoe Box Principle, is useful in answering questions such as: “ Is there an item having a given property? “ When successfully applied, the principle tells us only that object exists. The principle will not help us find the object or determine how many objects there are.

Pigeonhole Principle (First Form)

- If n pigeons fly into k pigeonholes and **k** < **n**, some pigeonhole contains at least two pigeons.
- The truth of this statement can be established by contradiction. In the conclusion is false, that some pigeonhole contains at least two pigeons, then each pigeonhole contains at most one pigeon. Then there can only be at most **k** pigeons. But the hypothesis states that there are **n** pigeons and **n** > **k**, hence, we have a contradiction.

Pigeonhole Principle (First Form)

Example 1:

Ten persons have first names Alice, Bernard, and Charles and last names Lee, McDuff, and Ng. Show that at least two persons have the same first and last names.

Solution:

There are nine possible names for the 10 persons. If we think of the persons as pigeons and the names as pigeonholes, we can consider the assignment of names to people to be that of assigning pigeonholes to the pigeons. By the Pigeonhole Principle some name (pigeonhole) is assigned to at least two persons (pigeons).

Pigeonhole Principle (Second Form)

- If f is a function from a finite set X to a set Y and $|X| > |Y|$, then $f(x_1) = f(x_2)$ for $x_1, x_2 \in X$, $x_1 \neq x_2$
- The second form of the Pigeonhole Principle can be reduced to the first form by letting X be the set of pigeons, and Y be the set of pigeonholes. We assign pigeon x to pigeonhole $f(x)$. By the first form of the Pigeonhole Principle, at least two pigeons, $x_1, x_2 \in X$, are assigned to the same pigeon, that is, $f(x_1) = f(x_2)$ for some $x_1, x_2 \in X$, $x_1 \neq x_2$.

Pigeonhole Principle (First Form)

Example 2:

Show that if we select 151 distinct computer science courses numbered between 1 and 300 inclusive, at least two are consecutively numbered.

Solution:

Let the selected course numbers be

$$c_1, c_2, \ldots, c_{151}.$$

Consider the courses $c_j = c_i + 1$. These are the courses

$$c_1 + 1, c_2 + 1, \ldots, c_{151} + 1$$

Pigeonhole Principle (Second Form)

Example 2:

Show that if we select 151 distinct computer science courses numbered between 1 and 300 inclusive, at least two are consecutively numbered.

Solution (cont.):

These 2 sets of courses make up 302 numbers with value ranging between 1 and 301 (because 300 is the highest course number). Since there are only 300 courses, by the second form of the Pigeonhole Principle, at least two of these 302 values coincide. Since the numbers in the first set are given to be all distinct, hence, the numbers in the set are also distinct. It must then be that one of the numbers in the first set and one of the second set are equal. Thus we have

$$c_i = c_j + 1, \quad \text{for some } i \text{ and } j$$

which implies that c_i follows c_j, and thus at least 2 courses are consecutively numbered.

Pigeonhole Principle (Second Form)

Example 3:

An inventory consists of a list of 80 items, each marked "available" or "unavailable." There are 45 available items. Show that there are at least two available items in the list exactly nine items apart. (For example, available items at positions 13 and 22 or positions 69 and 78 satisfy the condition.)

Let a_i denote the position of the ith available item. We must show that $a_i - a_j = 9$ for some i and j. Consider the numbers

$$a_1, a_2, \ldots, a_{45}$$

and

$$a_1 + 9, a_2 + 9, \ldots, a_{45} + 9$$

The 90 numbers that these 2 sets make up have possible values only from 1 to 89 (since 80 is the highest inventory number). By the second form of the Pigeonhole Principle, two of the numbers must coincide. We cannot have two from the same set identical (the 45 items available are uniquely numbered). Thus some number from the first set is equal to some number in the second set. Therefore, $a_i - a_j = 9$ for some i and j, as desired.

-- End of Presentation --